Lemma D'. Let V be a vector space over K and suppose that $V = sp(v_1, \dots, v_n)$ for some $v_1, \dots, v_n \in V$. Let $u_1, \dots, u_m \in V$ be lin. indep. Then $m \leq n$ and moreover one can delete m vectors from the list $v_1, \dots, v_n$ s.t. the remaining list, say $v_1', \dots, v_{n-m}'$ together with $u_1, \dots, u_m$ still span V, i.e.

$$sp(u_1, \dots, u_m, v_1', \dots, v_{n-m}') = V.$$

Proof of the "moreover" part of Lemma D'.

In the proof of Lemma D we have already seen that $m \leq n$. Now apply the procedure described in the proof of Lemma D for steps $j = 1, 2, \dots, m$. After m steps we arrive at a list $u_1, \dots, u_m, v_1', \dots, v_{n-m}'$ of the type claimed by Lemma D'.

See also Remark on the last page of lecture "11.basis.a".

$\blacksquare$

Thm. Let V be a finite dimensional vect. space over K. Then V has a basis with finitely many elements. Moreover, every basis of V is finite and has the same number of elements.

⊛ This thm is true even if V is not finite dimensional (called "Hamel basis theorem"). But in this course we will only prove the Theorem for finite-dimensional V. The case when V is infinite dimensional proven in the course "Grundstrukturen" in 2'nd semester.

For the proof of the Theorem we need some preparation.

Lemma E. Let $w_1, \dots, w_\ell \in V$ and assume that $w_j \notin Sp(w_1, \dots, w_{j-1})$ $\forall$ $1 \le j \le \ell$. Then the list $w_1, \dots, w_\ell$ is linear independent.

Proof. If $w_1, \dots, w_\ell$ were linearly dependent, then by Lemma A $\exists$ $1 \le j \le \ell$ s.t. $w_j \in Sp(w_1, \dots, w_{j-1})$.
Contradiction. ∎

Lemma F. Suppose that $Sp(v_1, \dots, v_n) = V$. Then $\exists$ a subset of $\{v_1, \dots, v_n\}$ which is a basis for V.

Proof. The proof is algorithmic & is based on n steps. At each step we'll consider another vector from the list $v_1, \dots, v_n$ and decide whether or not to drop it from the list. At step #j we consider v_j

Step 1. If $v_1 = 0$ we drop v_1 from the list.

If $v_1 \neq 0$ we keep it.

Step $2 \leq j \leq n$. If $v_j \in Sp(v_1, \dots, v_{j-1})$ we drop v_j. Otherwise we keep it.

After performing n steps as above we obtain a new, possibly shorter, list: $v_{i_1}, v_{i_2}, \dots, v_{i_m}$, where $1 \leq i_1 < i_2 < \dots < i_m \leq n$.

We claim that $Sp(v_{i_1}, \dots, v_{i_m}) = V$.

Indeed, in any of the steps $1 \leq j \leq n$ we droped the vector v_j only if $v_j \in Sp(v_1, \dots, v_{j-1})$.

$\Rightarrow Sp(v_{i_1}, \dots, v_{i_m}) = Sp(v_1, \dots, v_n) = V$.

We also claim that $v_{i_1}, \dots, v_{i_m}$ is lin. independent.

Indeed $\forall\ 1 \leq k \leq m$, we have $v_{i_k} \notin Sp(v_1, v_2, \dots, v_{i_k - 1})$

$\Rightarrow v_{i_k} \notin Sp(v_{i_1}, v_{i_2}, \dots, v_{i_{k-1}})$.

By Lemma E, $v_{i_1}, v_{i_2}, \dots, v_{i_m}$ is lin. independ.

Together with the fact that $Sp(v_{i_1}, \dots, v_{i_m}) = V$ we get that $v_{i_1}, \dots, v_{i_m}$ is a basis for V. $\blacksquare$

We can now prove the Thm.

Proof of the Thm. The Thm. states three things:

1. V has a basis with finitely many elements.
2. Every basis of V has finitely many elements.
3. Every two bases $\mathcal{A}$ and $\mathcal{B}$ are finite and moreover $|\mathcal{A}| = |\mathcal{B}|$.

Statement 1: follows from Lemma F. Take any finite set S s.t. $Sp(S) = V$ (such an S exists b.c. by assumption V is finite dimensional).
By Lemma F, a subset $S' \subseteq S$ is a basis for V.

Statement 2: every basis $\mathcal{C}$ of V is finite.
Suppose by contradiction that $\mathcal{C}$ is a basis of V but $\mathcal{C}$ is not a finite set. Let S be a finite basis of V (such S exists by assumption) say $S = \{v_1, \dots, v_n\}$. Choose $n+100$ vectors $u_1, \dots, u_{n+100} \in \mathcal{C}$. They are linearly independent.
By Lemma D, $n+100 \leq n$. Contradiction!
This shows that every basis $\mathcal{C}$ of V is finite.

Statement 3: let $\mathcal{A}, \mathcal{B}$ be two bases of V.

Then $|\mathcal{A}| = |\mathcal{B}|$.

Indeed, by statement 2, $|\mathcal{A}|, |\mathcal{B}| < \infty$.

Now, $Sp(\mathcal{B}) = V$, and $\mathcal{A}$ is lin. independent,

so by Lemma D we get that $|\mathcal{A}| \leq |\mathcal{B}|$.

But also $Sp(\mathcal{A}) = V$ and $\mathcal{B}$ is lin. independent,

so by Lemma D again we get $|\mathcal{B}| \leq |\mathcal{A}|$.

$\Rightarrow |\mathcal{A}| = |\mathcal{B}|$. ▨

Def. Let V be a finite dimensional vect.space over K. We define the dimension of V to be the number $n \in \mathbb{Z}_{\geq 0}$ of elements in a basis (any basis!) of V. We write $\dim V := n$.

In case V is not finite dimensional we write $\dim V = \infty$.

(*) $\dim V$ is well defined!

(the definition does not depend on a specific choice of a basis for V).

Remark. $\dim \{0\} = 0$. B.c. ϕ is a basis for the space $\{0\}$, and $|\phi| = 0$.

Examples.

1) $\dim K^n = n$.

2) $\dim K[x]_d = d+1$.

3) $\dim M_{m\times n}(K) = m\cdot n$.

4) Let S be a set, recall that $K^S := \{f \mid f: S \to K\}$ is a vector space over K. We have $\dim K^S = |S|$.

If S is a finite set, say $S = \{s_1, \dots, s_n\}$, find a basis for K^S. (when S is infinite, we don't have a nice basis.)